Roll no : 241501199
Name : Sherwin Richie J

CS23532-COMPUTER NETWORKS-LAB MANUAL

PRACTICAL – 9

AIM:

To design and implement multiple subnets in Cisco Packet Tracer, assign static IP addresses to devices, connect the subnets using a router, and simulate packet transmission to observe communication with and without a router.

PROCEDURE:

1. **Create the Network Topology:**
   - Open **Cisco Packet Tracer**.
   - Place the following devices:
   - 1 Router — Cisco 2911
   - 2 Switches — Cisco 2960
   - 4 PCs
   - Connect the devices

2. **Create the subnets:**
   The network 192.168.10.0/24 was divided into two /25 subnets.

   Subnet 1

- Network Address: 192.168.10.0
- Subnet Mask: 255.255.255.128
- Usable IP Range: 192.168.10.1 – 192.168.10.126
- Broadcast Address: 192.168.10.127

Subnet 2

- Network Address: 192.168.10.128
- Subnet Mask: 255.255.255.128

- Usable IP Range: 192.168.10.129 – 192.168.10.254
- Broadcast Address: 192.168.10.255

3. **Assign Static IP Addresses:**

   Configure the PCs through:

   PC → Desktop → IP Configuration

4. **Configure the Router:**

   Open:

   Router0 → CLI

   Command prompt:

   enable

   configure terminal

   interface gigabitEthernet 0/0

   ip address 192.168.10.1 255.255.255.128

   no shutdown

   exit

   interface gigabitEthernet 0/1

   ip address 192.168.10.129 255.255.255.128

   no shutdown

   exit

   end

   show ip interface brief

   Outcome:

   GigabitEthernet0/0 192.168.10.1 up up

   GigabitEthernet0/1 192.168.10.129 up up

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5. **Test Communication Between Different Subnets:**

   Open:

   PC0 → Desktop → Command Prompt

   Command:

   ping 192.168.10.130

   Outcome:

   Packets: Sent = 4

   Received = 4

   Lost = 0

6. **Test Communication Within Subnet 2:**

   Open:

   PC2 → Desktop → Command Prompt

   Command:

   ping 192.168.10.140

   Outcome:

   Packets: Sent = 4

   Received = 4

   Lost = 0

7. **Simulate Packet Transmission:**
   - Switch from Realtime mode to Simulation mode.
   - Select Add Simple PDU.
   - Click PC0 and then PC2.

- Click Auto Capture / Play.
- Observe the packet travelling through:
- PC0 → Switch0 → Router0 → Switch1 → PC2

8. Test Communication Without the Router:

- Return to **Realtime** mode.
- Disconnect the cable between **Switch0 and Router0**.
- Do not delete the router or switches.
- From **PC0 → Desktop → Command Prompt : ping 192.168.10.130**

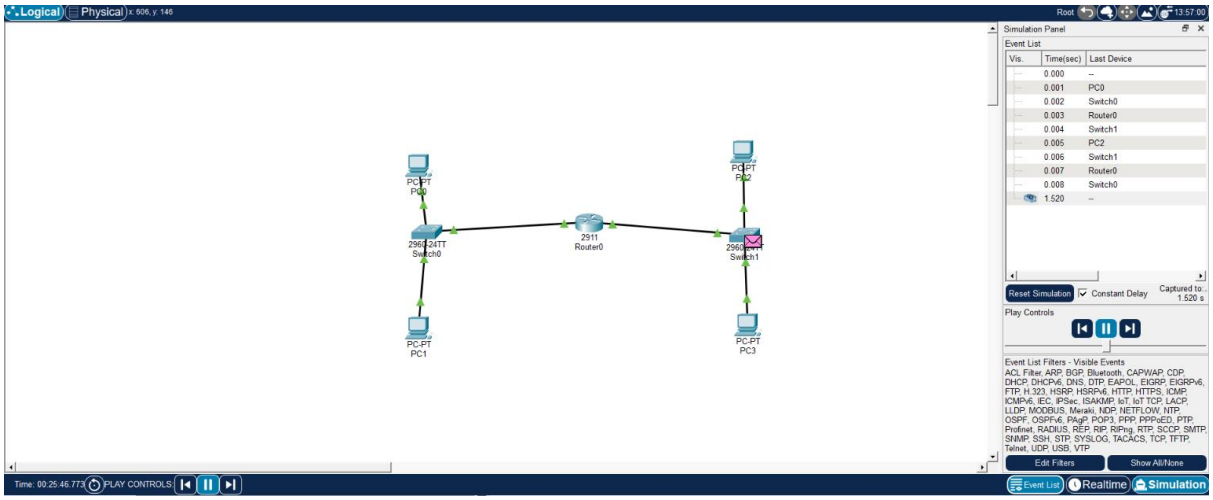
Outcome:

Packets: Sent = 4

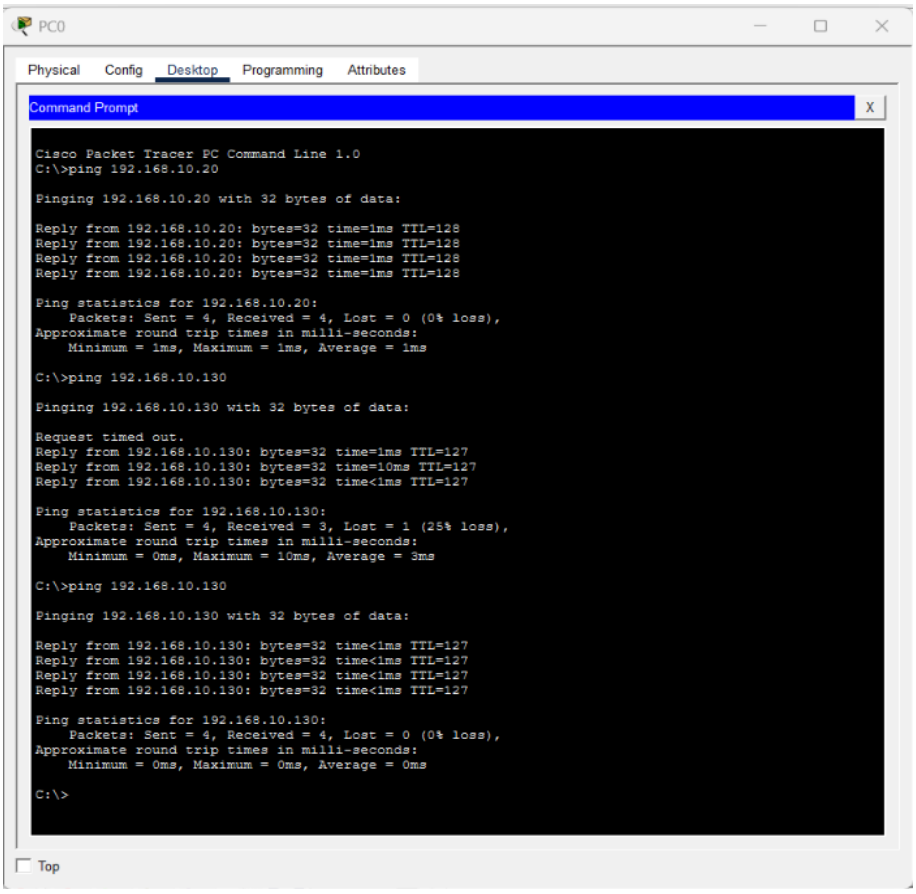
Received = 0

Lost = 4

SCREENSHOTS:



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Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.128
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

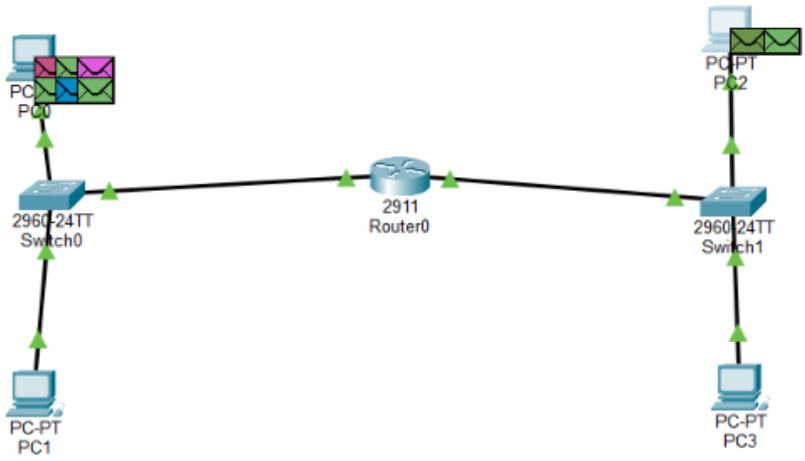
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.10.129 255.255.255.128
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     192.168.10.129  YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
Router#
```

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RESULT:

Thus, the experiment successfully demonstrated subnetting, static IP addressing, router-based inter-subnet communication, and the importance of a router for communication between different subnets.